

Seminar Quiz 5

The Economy

Sciences Po Paris - Reims

15 minutes. No extra papers. You are free to use calculators.

Exercise 1

Consider an economy with taxes in the Keynesian model:

$$Y = c_0 + c_1 Y_d + I + G$$

Government spendings and investments add up to \$800, fixed consumption is \$300 and the collected tax is \$200. We also know that the total GDP amounts to \$2,000.

1. Calculate the multiplier for this economy (we assume taxes are proportional to income i.e. $T = tY$).
(3 points)

Finding c_1 :

$$Y = c_0 + c_1 Y_d + I + G$$

$$2000 = 300 + c_1 Y_d + 800$$

$$900 = c_1 Y_d$$

$$\text{since, } Y_d = Y - T \rightarrow Y_d = 2000 - 200 = 1800$$

$$\text{hence, } c_1 = 0.5$$

Since taxes are proportional to income i.e. $T = tY$, we can plug in T and Y to find that $t = 0.1$. Then the multiplier with taxes is:

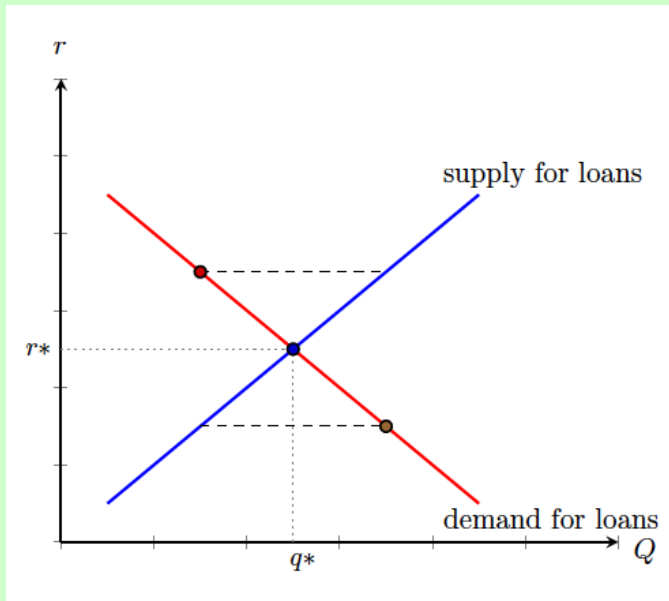
$$\begin{aligned} \frac{1}{1 - c_1(1 - t)} &= \frac{1}{1 - 0.5(1 - 0.1)} \\ &= \frac{1}{0.55} \end{aligned}$$

2. Draw the aggregate demand for this economy along with the $Y = AD$ line. Show what would happen if the MPC were to decrease. Show how this would affect the multiplier using the formula. (2 points + 1 point for good labeling)

The AD line pivots downwards hence the slope (c_1) decreases. See **PS8** solutions for the increase in MPC graph.

Exercise 2

1. Consider an economy where there is high inflation. The central bank wants to intervene and control the inflation. What type of monetary policy would they follow? Draw a supply and demand graph for the loan market to explain your reasoning. (2 points)



The government follows a contractionary monetary policy to withdraw money out of the market. This "tightening" measure pushes people to save more and spend less, decreasing inflation.

- The S curve will shift **left** (remember down for decrease in prices, left for quantities) as a result of the contractionary monetary policy.
→ increasing r and decreasing q

2. Steven wants to attend college but he doesn't have money to pay for the tuition, which is \$10,000. He decides to take on a student loan for two years. If the interest rate on this loan is 10%, how much will he have to pay back in **total** after two years? (2 points)

Recalling the FV formula for an amount of money that is being borrowed today:

$$FV = PV(1 + r)^n$$

$$FV = 10,000(1 + 0.1)^2$$

$$FV = 12,100$$